Shourya Bhardwaj

AI Systems & Agentic Software Builder

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Builder focused on local-first AI agents, automation, and biologically inspired simulation. I work from architecture through implementation, debugging, validation, and iteration to turn ambitious ideas into working systems.

SELECTED ENGINEERING PROJECTS

Nova

In development

Primary builder | AI Agents / Systems Engineering

- Architected a staged request pipeline - normalization, routing, planning, validation, execution, and verification - so model proposals remain inspectable before they become computer actions.
- Built across a C++ runtime and Python voice/automation components, integrating OpenWakeWord, faster-whisper, filesystem primitives, and Playwright-backed browser operations.
- Added content-aware file creation, one bounded repair attempt, and postcondition readback verification; reported regression results passed 27/27 Branch 3, 15/15 mapper, and 15/15 decomposition tests after the fix.

github.com/shoyo-nova92/Nova-AI

Neurofly

Experimental

Project builder | Computational Neuroscience / Simulation

- Developing an interactive sensing-to-movement simulation inspired by Drosophila olfaction, mapping odor concentration fields through receptor/neural responses into navigation behaviour.
- Designing the arena for controllable odor sources, bilateral sensing, inspectable neural activity, and competing stimuli so behaviour can be studied rather than pre-scripted.
- Defined a future hardware path around an ESP32-based robot intended to mirror the virtual neural-control loop; physical integration remains in progress.

FREELANCE & CREATIVE EXPERIENCE

Independent Design & Editing

Visual design | Video editing | Digital communication

- Produced promotional and product-focused visual compositions, including selected work for LinkScope, Identix, Clarion AI, and Spectra AI.
- Worked across visual design and video editing, focusing on hierarchy, clarity, and polished presentation for digital audiences.

TECHNICAL ACTIVITY

OpenAI / Codex Hands-On Workshop

Oct 2026

- Practised structured prompting, AGENTS.md project rulebooks, parallel specialist agents, independent QA/review, and full-stack deployment concepts.
- Applied the multi-agent builder/checker workflow to ongoing Nova development and to the construction of this portfolio.

TECHNICAL CAPABILITIES

AI & agent systems
Planning & tool use; validation/execution workflows; voice-input pipelines; browser & desktop automation

Applied interaction
faster-whisper; OpenWakeWord; Playwright; input normalization & routing

Creative
Visual design; video editing; visual communication

Systems & software
C++; Python; CMake; runtime architecture; filesystem operations; debugging & testing

Research & simulation
Biologically inspired control; olfactory simulation; neural navigation exploration